

The Fertility, Marriage, and Gender Equality Quandary

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Motivation

- Three trends underpin the grand gender convergence in the last century:
 1. [Falling fertility](#) (Guinnane 2011)
 2. [Declining dual parenthood](#) (Stevenson and Wolfers 2007)
 3. [Converging gender \(income\) gaps](#) (Goldin 2014)
- Many policies target individual outcomes
- Existing research often treats them in isolation or pairwise, lacking a unified framework

This Paper

1. Document a **three-way trade-off**: simultaneously achieving high fertility, low single parenthood, and gender income equality is statistically rare
 $\implies$ deep dependency structure & potential policy boundaries
2. Develop a **unified equilibrium model** of marriage, fertility, and female labor supply to rationalize the facts and evaluate policy
3. Calibrate to Mexico (1990–2015); decompose drivers and test out-of-sample predictions

Key Findings

1. Mutual information analysis shows that pairwise analysis misses **synergy**
2. Policies targeting *levels* (wages, marriage, childcare cost) cannot resolve the trade-off, but targeting the **distribution of childcare** can—under two transparent conditions
3. Mexico 1990–2015:
 - Gender-neutral TFP explains ~50% of declining fertility and dual parenthood
 - Gender-biased TFP and education reversal drive income convergence
 - 2023 out-of-sample fertility forecast: **1.93** vs. actual **1.91**
4. Dynamic propagation through gendered impacts of single parenthood

Literature

- Pairwise studies of fertility, marriage, and gender gaps (Erosa et al. 2010, Regalia & Ríos-Rull 2001, Stevenson & Wolfers 2007)
→ Establish three-way trade-off and provide explanation
- Gender-biased technical change (Greenwood et al. 2005, Ngai & Petrongolo 2017)
→ Show gender-neutral TFP also drives convergence
- Boys' disadvantage in single-parent families (Autor et al. 2019)
→ Embed in macro model with dynamic propagation

Roadmap

- Empirical findings
- Model
- Policy analysis
- Quantitative analysis
- Dynamic extension
- Conclusion

Motivating Facts

Empirical Approach

1. Assemble **four datasets** with fertility, dual parenthood, and gender income equality
2. Create dummy variables at sample medians:
 - High fertility (F)
 - Dual parenthood (M)
 - Gender income equality (G)
3. Visualize intersections using Venn diagrams
4. Compute mutual information to characterize dependency structure

Dartboard benchmark: under independence, trinity share is $0.5^3 = 12.5\%$

OECD Sample

37 OECD countries, 1970–2014

721 observations

- Total fertility rate (1.69)
- Out-of-wedlock birth share (31.4%)
- Gender median earnings gap (17.2%)

Advantage: longest horizon

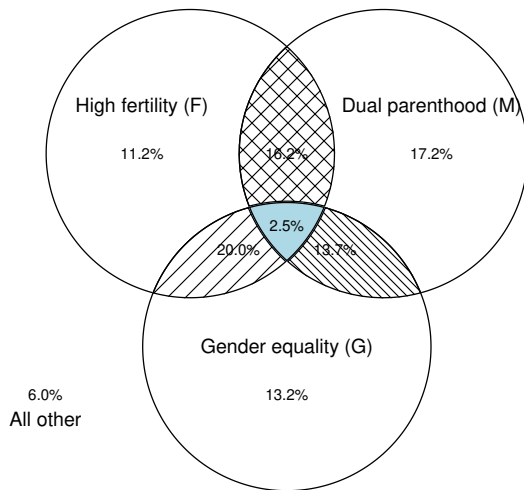
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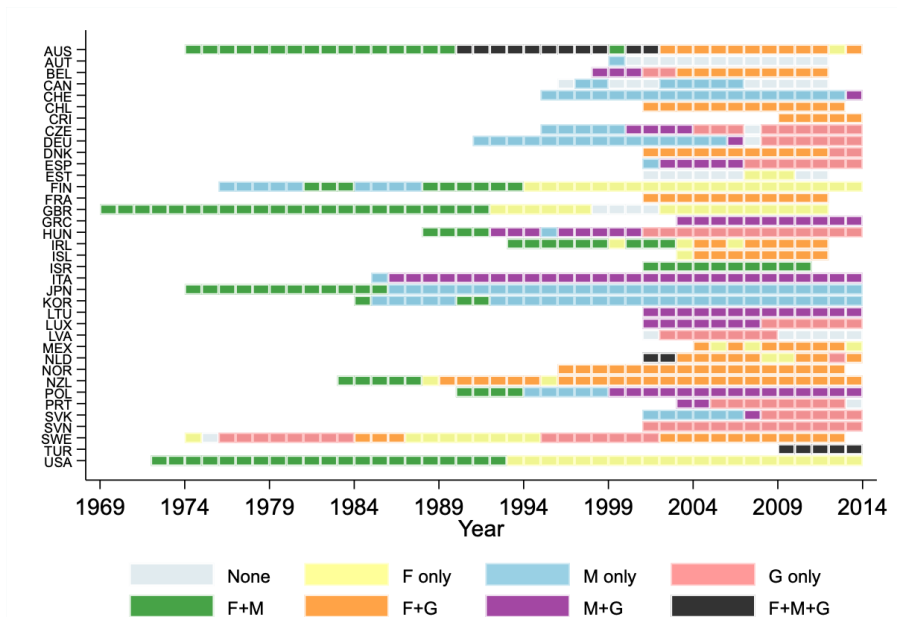
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Trinity share: 2.5%

MI analysis: synergy

OECD Sample – Transition Paths



World Sample

95 countries, 1990–2015

1,130 observations

- Total fertility rate (2.81)
- Children with both parents (75.9%)
- Female labor income share (32.3%)

Advantage: better measurement of family structure, wider coverage

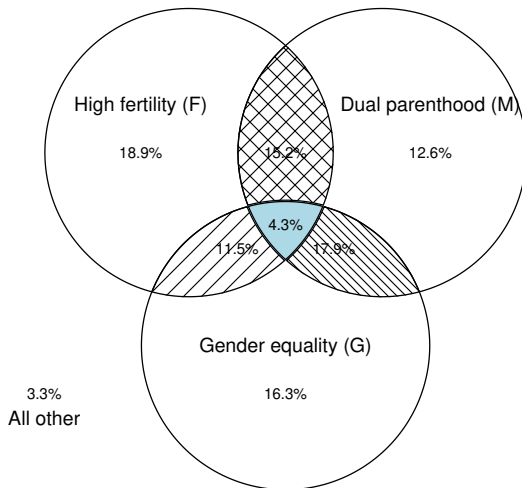
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Trinity share: 4.3%

MI analysis: synergy

U.S. Samples

map

States: 50 + D.C., 2023

- Births per 1000 women
(56.9)
- Single-mother share (17%)
- Gender earnings gap
(25.6%)

Commuting Zones:

741 CZs, 2000

Advantage: uniform institutional setting

U.S. Samples

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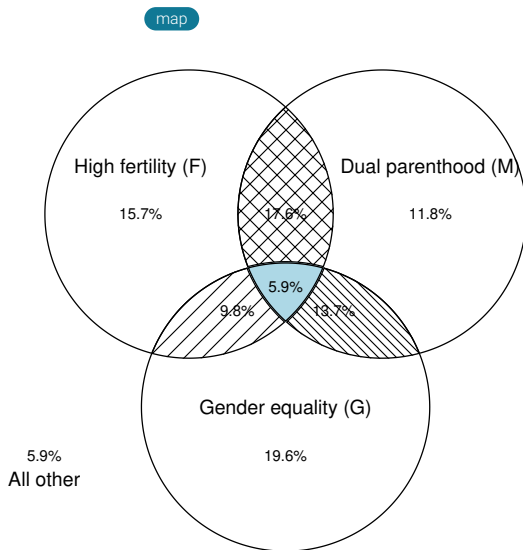
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Commuting Zones:

741 CZs, 2000

Trinity share: 7.6%

Advantage: uniform institutional setting



States trinity share: 5.9%

MI analysis: synergy

Mutual Information: Synergy, not Redundancy

- **Synergy**: conditioning on any one variable *strengthens* the dependence between the other two
- Implication: pairwise analyses miss joint structure

Sample	Total Corr.	Interaction	Interpretation
OECD	0.165	−0.035***	Synergy
World	0.169	−0.026***	Synergy
U.S. States	0.112	−0.007	Synergy

Example (OECD): $I(F; M) = 0.043$ bits, but $I(F; M | G) = 0.078$ bits—**nearly double**

Questions

1. What explains the three-way trade-off?
2. Are there policies that can escape the trade-off?
3. What role does technological progress play?

Theory

Basic Setup

- Building on Becker (1973), Choo and Siow (2006)
- Individuals with gender $g \in \{\sigma, \varphi\}$ and preference

$$u(c, n) = \left((1 - \beta) \cdot c^{\frac{\rho-1}{\rho}} + \beta \cdot n^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (1)$$

where $\rho > 1$ so that $u(c, 0)$ is well-defined

- Homogeneous wage within gender: w^{σ} and w^{φ}
- Gender wage gap: $\Gamma^w = w^{\sigma} / w^{\varphi}$

Marriage and Fertility – Men

- If single, men consume labor income but have no children

$$V^{\sigma^{\uparrow},s} = u(w^{\sigma^{\uparrow}}, 0) \quad (2)$$

- Once married, husbands transfer α share of income to wives

$$V^{\sigma^{\uparrow},m} = u((1 - \alpha)w^{\sigma^{\uparrow}}, n^m) \cdot \lambda \quad (3)$$

- α is endogenous in marriage market equilibrium
- λ is the psychic benefit of marriage

Marriage and Fertility – Women

- Single women solve

$$V^{\text{♀},s} = \max_{c,l,n} u(c, n) \quad \text{s.t.} \quad c = w^{\text{♀}}l, \quad l = 1 - \chi n \quad (4)$$

- Married women solve

$$V^{\text{♀},m} = \max_{c,l,n} u(c, n) \cdot \lambda \quad \text{s.t.} \quad c = \alpha w^{\text{♂}} + w^{\text{♀}}l, \quad l = 1 - \chi n \quad (5)$$

- Trade-off between consumption and fertility through female labor supply
- In the baseline, wives bear all childcare. Results hold so long as $\chi^{\text{♂}} < \chi^{\text{♀}}$

Marriage Market Equilibrium

- Idiosyncratic taste shock ϵ for marriage, Fréchet with scale θ
- Marriage rates:

$$\mathcal{M}^{\sigma} = \frac{1}{1 + (V^{\sigma,s}/V^{\sigma,m})^{\theta}}, \quad \mathcal{M}^{\varphi} = \frac{1}{1 + (V^{\varphi,s}/V^{\varphi,m})^{\theta}} \quad (6)$$

- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- Two-dimensional price (α, n^m) : marriage bundles income-sharing *and* fertility-sharing

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- Equilibrium: $\mathcal{M}^{\sigma} = \mathcal{M}^{\varphi} = \mathcal{M}$
- **Two-dimensional price** (α, n^m) : marriage bundles income-sharing *and* fertility-sharing
- Lemma 1: For any $\{w^{\sigma}, w^{\varphi}\}$, there exists a unique equilibrium (α, n^m)
- Lemma 2: Gains from marriage: $V^{\varphi,m}/V^{\varphi,s} = \lambda \cdot (1 + \alpha\Gamma^w)$

Three-Way Trade-Off: Intuition

- Equilibrium conditions reveal fundamental tension:

$$\mathcal{M} = \frac{(\lambda \cdot (1 + \alpha \Gamma^w))^\theta}{1 + (\lambda \cdot (1 + \alpha \Gamma^w))^\theta} \quad (7)$$

$$\Gamma^y = \Gamma^w / l^\varnothing \quad (8)$$

$$l^\varnothing = 1 - \chi n \quad (9)$$

- Insight:** Higher n reduces $l^\varnothing$, requiring low Γ^w for gender income equality—but low Γ^w reduces $\mathcal{M}$

Traditional Policy Effects

Policy	Fertility (n)	Marriage ($\mathcal{M}$)	Gender Gap (Γ^y)
Pro-marriage ($\uparrow \lambda$)	+	+	+ (widens)
Gender equality ($\uparrow w^{\text{♀}}$)	−	−	− (narrows)
Reduce child costs ($\downarrow \chi$)	+	+	+ (widens)

- No *level* policy can improve all three outcomes
- Trade-off binds because wives' disproportionate childcare burden ties fertility to gender gaps through relative labor supply

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- No *level* policy can improve all three outcomes
- Trade-off binds because wives' disproportionate childcare burden ties fertility to gender gaps through relative labor supply
- What if the [allocation](#) of childcare is itself a policy lever?

Model with Husband-Provided Childcare

- Extension: $\chi^{\text{♀}}$ for wives, $\chi^{\text{♂}}$ for husbands
- $\chi = \chi^{\text{♀}} + \chi^{\text{♂}}$ for single mothers
- Wife's effective childcare cost: $\tilde{\chi}^{\text{♀}} = \frac{\alpha w^{\text{♂}} \chi^{\text{♂}} + w^{\text{♀}} \chi^{\text{♀}}}{\alpha w^{\text{♂}} + w^{\text{♀}}}$
- Consider redistributing δ childcare from wives to husbands

Redistributing Childcare to Husbands

empirical

- **Proposition:** Redistribution improves all three outcomes if:
 1. Wife's earnings exceed spousal transfers:

$$w^{\varnothing} > \alpha \cdot w^{\sigma} \quad (10)$$

2. Husband's gain from fertility exceeds consumption loss.

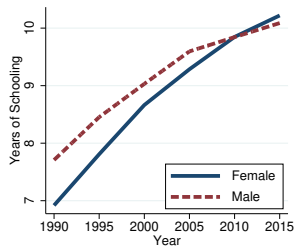
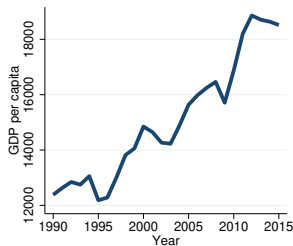
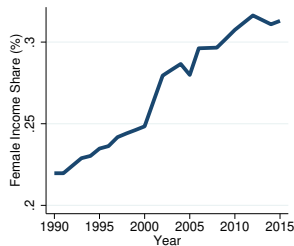
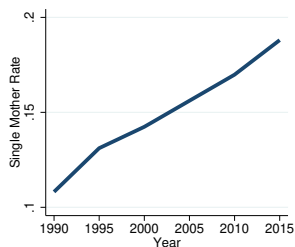
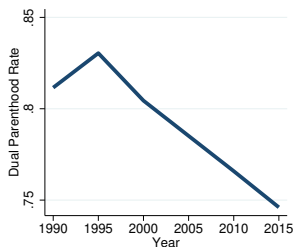
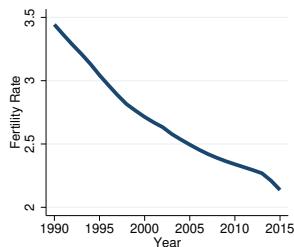
$$\frac{\beta}{1-\beta} \cdot \left(\frac{c^{\sigma,m}}{n^m} \right)^{\frac{1}{\rho}} > (1-\alpha)w^{\sigma} \cdot \left(\frac{n^m}{\partial n^m / \partial \delta} + \chi^{\sigma} \right), \quad (11)$$

More likely if: β is large, initial χ^{σ} is small, and fertility response is strong

- When both conditions hold, redistribution is **Pareto-improving**

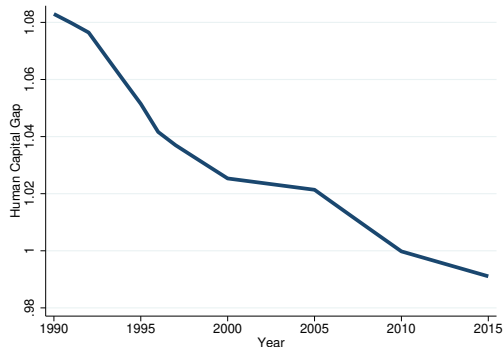
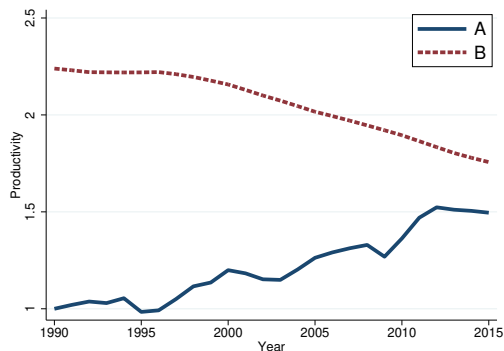
Quantitative Analysis

Mexico's Transition (1990–2015)



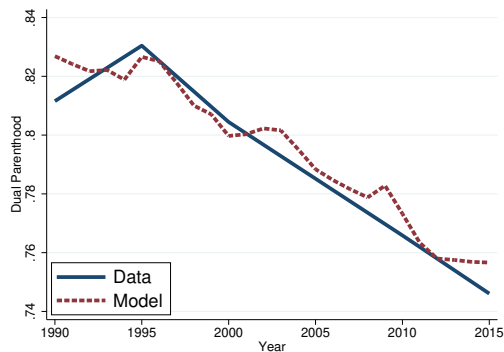
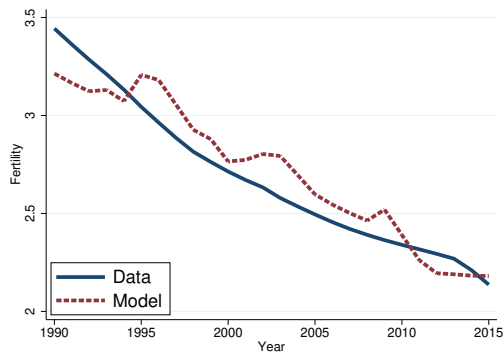
Calibration

- Decompose wages into three structurally interpretable components:
 - $A_t = w_t^{\sigma}$: gender-neutral productivity
 - Γ_t^h : gender gap in human capital
 - $B_t = (w_t^{\sigma} / w_t^{\varphi}) / \Gamma_t^h$: gender-biased productivity



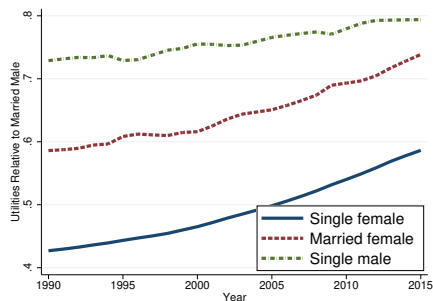
Model Fit

- Fix $\chi = 0.075$ (de La Croix and Doepke 2003), assuming $\chi^{\sigma^*} \approx 0$
- Calibrate: $\beta = 0.104$, $\rho = 1.50$, $\lambda = 1.03$, $\theta = 3.40$

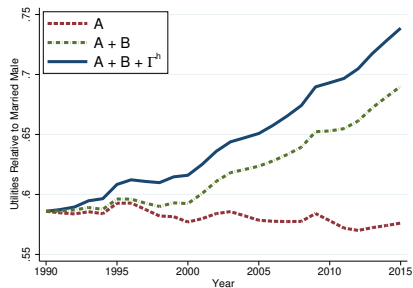
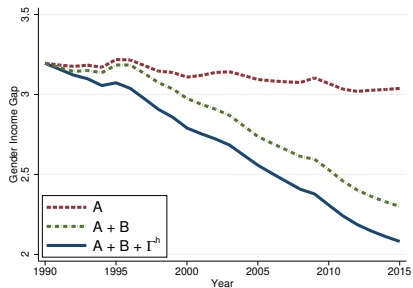
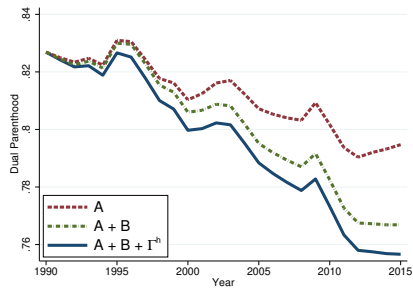
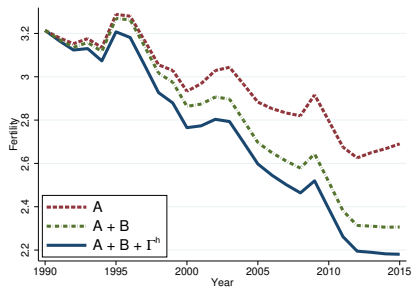


Equilibrium Outcomes

- α_t falls only modestly despite sharp decline in n^m : narrowing Γ^w requires larger share of smaller relative income
- Husband-wife welfare gap shrinks from 42% to 26% (driven by rising w^{f} , not larger transfers)
- $\alpha \approx 0.10$ satisfies condition (1); condition (2) also holds throughout $\Rightarrow$ redistribution would be Pareto-improving

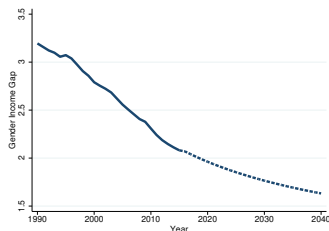
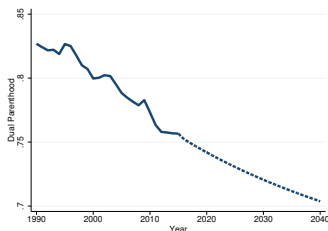
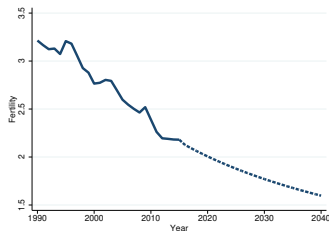
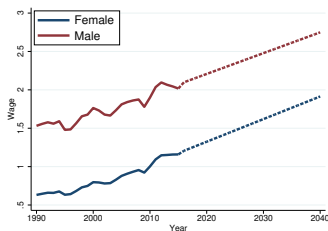


Decomposition



Model-Based Prediction

- Extrapolate wages linearly from 1990–2015 trend
- 2023 out-of-sample: model predicts fertility of 1.93; actual is 1.91



Dynamic Extension

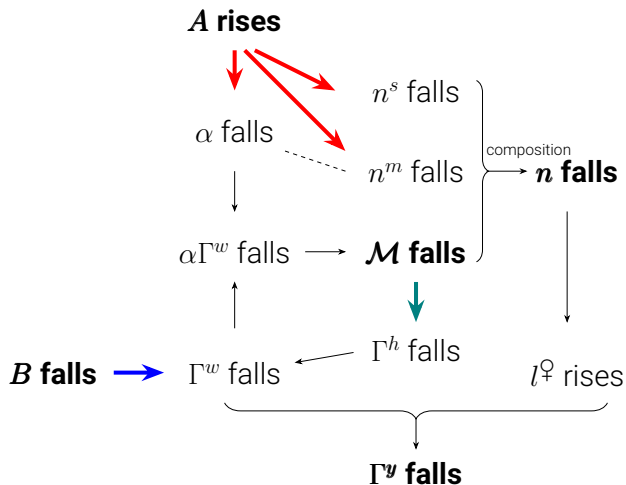
Endogenous Gender Human Capital Gap

- Single parenthood affects boys and girls differently
- Evidence: Bertrand & Pan 2013, Autor et al. 2019, Wasserman 2020, Reeves 2022, Autor et al. 2023
- Law of motion:

$$\Gamma_{t+1}^h = \mathcal{H}(\mathcal{M}_t) \quad \text{where } \mathcal{H}'(\mathcal{M}_t) > 0 \quad (12)$$

- Creates self-reinforcing feedback: declining marriage $\rightarrow$ lower male human capital $\rightarrow$ further marriage decline

Dynamic Mechanism



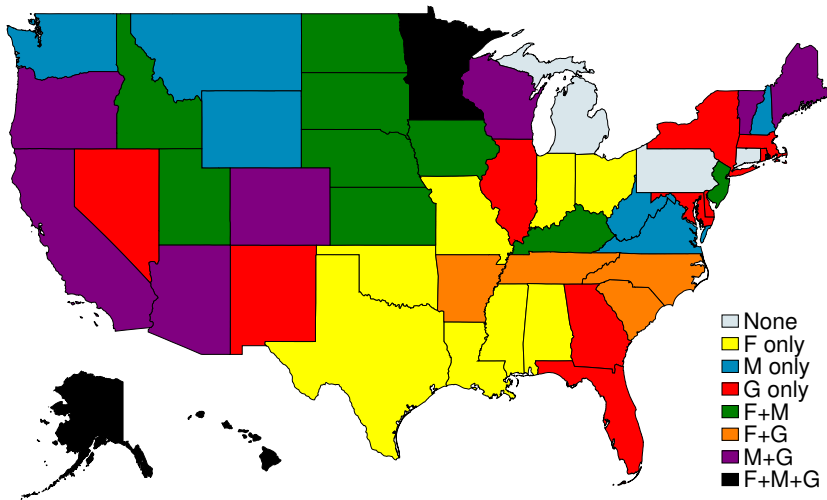
Red: A_t rises Blue: B_t falls Teal: Dynamic feedback

Conclusion

- A three-way trade-off with synergistic dependency structure
- Unified framework with two-dimensional marriage-market pricing
- Level policies trade one outcome against the others, while redistribution of childcare can loosen the trade-off under checkable conditions
- Mexico: both gender-neutral and gender-biased technological change matter; strong out-of-sample fit
- Dynamic propagation through male human capital amplifies effects across generations

U.S. States – Geographic Distribution

[back](#)



Three states achieve the trinity: [Alaska](#), [Hawaii](#), [Minnesota](#)

Organizing the Paternity-Leave Evidence

[back](#)

- **Gender-gap channel (robust)**: reserved quotas narrow within-household division of childcare in Spain, Germany, Norway, Quebec
- **Fertility channel (heterogeneous)**:
 - Null/negative in Spain, Norway (high-wage, low-fertility; modest β , higher baseline χ^{σ^*})
 - Positive in Quebec, low-baseline χ^{σ^*} subgroups in Germany
- Pattern tracks condition (2) of the proposition
- **Falsifiable prediction**: in middle-income settings (high β , low χ^{σ^*} , moderate w^{φ}), reforms should raise fertility *and* narrow gender gaps